V Muhammed Saad Sabeel

+91 6360487938 | mohammedsaadsabeel@gmail.com | Yelahanka, Bengaluru, Karnataka - 560064

LinkedIn | GitHub

Professional Summary

Electronics & Telecommunication Engineering student with a deep-rooted interest in electronics and hardware development. A naturally curious tech enthusiast who brings creativity and problem-solving to every project. Always eager to experiment with new technologies, adapt to new challenges, and continuously evolve as an Engineer.

Education

BMS Institute of Technology & Management — Bengaluru, India
Bachelor of Engineering in Electronics & Telecommunication Engineering, CGPA: 9.21 — *2023 – 2027*

Agragami PU College — Bengaluru, India
Class 12 Science, Percentage: 89% — *2022 – 2023*

Projects

Diabreeze: Non-Invasive Biomarker Detection System | *ESP32, Data Processing*

- Built a portable diagnostic system that processes real-time breath acetone telemetry to estimate blood glucose levels.
- Engineered an ESP32 data pipeline to instantly compute and visualize physiological and environmental metrics.

EVSE Charge Controller Simulator | *ESP32, FreeRTOS, Embedded C*

- Built an EVSE charge controller simulator using a FreeRTOS task scheduler to manage concurrent charging states, sensor reading, and safety protocols.
- Used hardware interrupts and binary semaphores for rapid fault detection, and generated dynamic PWM signals to simulate IEC 61851 communication.

QNX Deterministic Security Monitor | *C, QNX RTOS, Microkernel Architecture, IPC*

- Designed a security monitor on a QNX Microkernel using strictly isolated C programs to mitigate hardware side-channel attacks.
- Developed a high-priority mitigation engine triggered by IPCs to instantly freeze compromised processes mid-instruction and prevent data theft.

Smart Switching Using IR Remote and Mobile | *NodeMCU ESP32, IoT, Blynk*

- Developed a home automation system using a 4-channel relay module for appliance control.
- Implemented dual control modes: an IR remote for local control and the Blynk IoT platform for mobile access.

IoT Intrusion Detection via Side-Channel Analysis | *Raspberry Pi, I2C, Python, Edge AI*

- Built a non-intrusive edge monitoring system that uses hardware power analysis to detect malware, bypassing the need for software logs.
- Created an I2C data pipeline to sample current consumption every 0.2 seconds, building a dataset to train real-time Edge AI classification models.

Extracurricular

AI Model Evaluator | *Outlier AI*

- Contributed 330+ hours of RLHF and SFT tasks to train enterprise LLMs, evaluating model outputs for factual accuracy and logical consistency.
- Performed quality evaluation of AI-generated outputs across Mathematics and multilingual domains.

Webmaster | *IEEE Circuits and Systems Society (CASS)*

- Managed and maintained the IEEE CASS student chapter website, ensuring timely updates of announcements, events, and technical content.
- Collaborated with the organizing team to design and publish event pages.

Technical Skills

Languages: C, Python, SQL
Embedded & OS: ESP32, Raspberry Pi, FreeRTOS, QNX Neutrino Microkernel
Tools & Software: Keil uVision, Git, VS Code, Xilinx Vivado, Synopsys